

31 Red blood cells may contain a molecule known as 2,3-bisphosphoglycerate (2,3-BPG).

When 2,3-BPG binds to haemoglobin a higher partial pressure of oxygen is needed to bring about 50% saturation of haemoglobin with oxygen.

Which statements about the effect of 2,3-BPG are correct?

- 1 2,3-BPG in red blood cells causes the oxygen dissociation curve to shift to the right.
- 2 The binding of 2,3-BPG to haemoglobin reduces the Bohr effect.
- 3 The binding of 2,3-BPG to haemoglobin lowers the affinity of the haemoglobin for oxygen.
- 4 When 2,3-BPG is absent, oxyhaemoglobin is less likely to unload oxygen.

A 1 and 2

B 1 and 3

C 2 and 3

D 3 and 4